Learning Local Feature Descriptions in 3D Ultrasound

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***Abstract*—Tools for automatic image analysis are gaining importance in the clinical workflow, ranging from time-saving tools in diagnostics to real-time methods in image-guided interventions. Over the last years, ultrasound (US) imaging has become a promising modality for image guidance due to its ability to provide volumetric images of soft tissue in real-time without using ionizing radiation. One key challenge in automatic US image analysis is the identification of suitable features to describe the image or regions within, e.g. for recognition, alignment or tracking tasks. In recent years, features that were learned data-drivenly provided promising results. Even though these approaches outperformed hand-crafted feature extractors in many applications, there is still a lack of feature learning for local description of three-dimensional US (3DUS) images. In this work, we present a completely data-driven feature learning approach for 3DUS images for usage in target tracking. To this end, we use a 3D convolutional autoencoder (AE) with a custom loss function to encode 3DUS image patches into a compact latent space that serves as a general feature description. For evaluation, we trained and tested the proposed architecture on 3DUS images of the liver and prostate of five different subjects and assessed the similarity between the decoded patches and the original ones. Subject- and organ-specific as well as general AEs are trained and evaluated. Specific AEs could reconstruct patches with a mean Normalized Cross Correlation of 0.85 and 0.81 at maximum in liver and prostate, respectively. It can also be shown that the AEs are transferable across subjects and organs, with a small accuracy decrease to 0.83 and 0.81 (liver, prostate) for general AEs. In addition, a first tracking study was performed to show feasibility of tracking in latent space. In this work, we could show that it is possible to train an AE that is transferable across two target regions and several subjects. Hence, convolutional AEs present a promising approach for creating a general feature extractor for 3DUS.**

***Index Terms*—Deep learning, Feature learning, Convolutional autoencoder, Medical ultrasound analysis**

I. Introduction

Ultrasound (US) imaging is an important tool for diagnosis and image-guided interventions to visualize soft tissue in real-time without using ionizing radiation. Furthermore, in contrast to CT and MRI, a US station is portable and does not need an especially provided room so it can be used in any environment. By using time-resolved three-dimensional US (3DUS), it is possible to not only obtain two-dimensional slices but volumetric images of the tissue in real-time. However, in contrast to CT and MRI, US images are strongly affected by artifacts like speckle noise or shadows that complicates image analysis. In addition, image quality depends on the user's experience and the device settings [1].

Today, US imaging is already used in a plethora of clinical fields such as cardiology, vascular imaging [2], prenatal examinations [3] or image-guided interventions such as needle guidance [4], robotic surgery [5] or radiation therapy [6], [7]. The main computer vision tasks performed on US images are object recognition or detection, segmentation and classification. For example, image features were detected in two-dimensional US slices and dictionaries were learned to classify the view of apical echocardiograms in [8]. Another classification task is performed in [9] to classify liver lesions with a neural network and detect image features. In [10], convolutional neural networks detected vessels in US slices in real-time. The task we focus on is tracking of target structures in abdominal ultrasound. Real-time tracking is a challenging task due to organ motion induced, for example by respiration or pulsation. Furthermore, analysis of high dimensional 3DUS volumes is computional expensive. To deal with the high dimensional 3DUS data, our previous work focused on the detection of image features in 3DUS volumes and using these to segment deformable targets for motion tracking in time-resolved 3DUS volumes (4DUS) [11].

Especially for image-guided interventions, real-time tracking systems that are able to interpret US images and identify the exact target position are required. In radiation therapy, for example, the tumor position and shape need to be known to optimally treat the tumor. There are several approaches for analyzing US images using machine learning [3] or deep learning algorithms [12]. Using classical machine learning approaches, only predefined features can be detected. In computer vision, a frequently used feature extractor is SIFT [13]. In [14] it is used for non-rigid registration of 3DUS images, but the computing time of this method is in the range of seconds, which is prohibitive for real-time applications like tracking. In [15], a

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DOI 10.1109/BIBE50027.2020.00059

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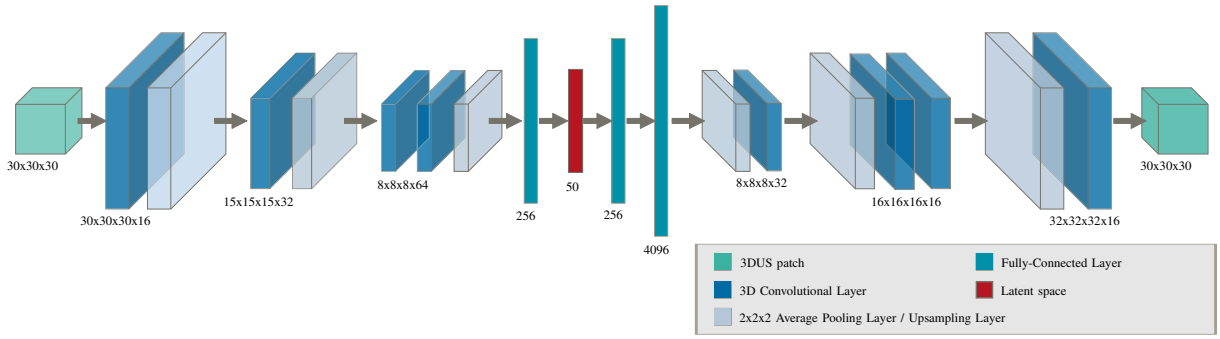


Fig. 1. Architecture of the AE.

real-time capable implementation of SIFT for 2D ultrasound is provided. However, due to the higher dimensionality of 3DUS volumes, the computing time would increase rapidly. Zheng et al. [16] showed that deep learning based convolutional neural networks (CNNs) can outperform SIFT in tasks such as object retrieval. With deep learning it is possible to extract nonlinear abstract features without predefining them, because a CNN learns the image features autonomously [12]. A disadvantage of deep learning approaches is that a large amount of data is needed and, if supervised learning methods are used, this data needs to be labeled. Due to the lack of available and suitable US images, developing deep learning based 3DUS tracking systems is challenging. To deal with that, a general feature extractor that is able to recognize the most important features in 3DUS volumes would be desirable.

In this paper, we present a new approach to compute features describing 3DUS volumes to reduce data dimensionality for tracking. A convolutional autoencoder (AE) with a custom loss function is used to compress patches of a 3DUS volume into a small latent space. In the latent space, the US patch is reduced to its most important features. These can then be used to analyze US volumes in latent space instead of analyzing the entire volume. In addition to tracking, it is possible to perform, for example, clustering or classification in the latent space. The AE is evaluated with respect to subject and organ generalizability. Additionally, an initial tracking study is performed to verify that AEs are usable for target tracking in 4DUS.

II. Methods

In this section, the AE architecture and its parameters, as well as the 4DUS scans used for training and testing, are introduced. Subsequently, the experiments for evaluating the AE and an initial tracking study are described.

A. Autoencoder Architecture

AEs are unsupervised learning networks consisting of an encoder that maps the input data $x \in \mathbb{R}^d$ into a latent space $\mathbb{R}^{d'}$ with $d' < d$ and a decoder that maps the data from the latent space onto a reconstruction $x^{(p)} \in \mathbb{R}^d$. Here, $d = N \cdot I \cdot J$ and N,I and J are the dimensionalities of the input data axes. The AE used here is a deep convolutional AE (illustrated in Fig. 1). It is trained to encode 3DUS patches with a size of 30×30×30 voxels into a small latent space. As can be seen in Fig. 1, the encoder has six layers and the decoder has seven layers. The encoder maps the patch into a low-dimensional ($d' = 50$) latent space and the decoder reconstructs the input patch. The activation function used in the AE is *PReLU* [17], except for the last Fully-Connected layer of the encoder and the decoder, where *tanh* and *sigmoid* are used for activation, respectively. The AE is trained for 100 epochs with a batch size of 32 and the *Adam* optimizer [18].

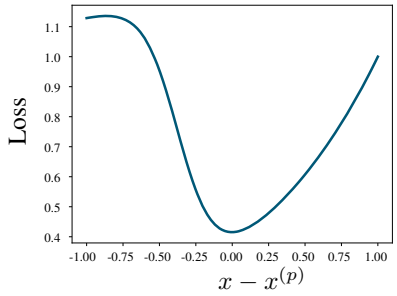


Fig. 2. The loss function *LUS* used for training. It penalizes stronger if a voxel is brighter in the decoded patch than in the original.

In abdominal US images, structures like big vessels appear darker than surrounding tissue. Because these structures are important landmarks for interpreting US images, they need to be reconstructed in a high accuracy. For this reason, a custom loss function, called *LUS*, is used defined by

$$\mathrm{LUS}(x, x^{(p)}) = \frac{1}{d} \sum_{i=0}^{d} \frac{1}{1 + e^{\alpha(x_i - x_i^{(p)}) + \beta}} + e^{(x_i - x_i^{(p)}) - 1}. \tag{1}$$

Here, x and $x^{(p)}$ are the input and decoded patches that are compared to each other and d is the number of voxels. In Fig. 2, the loss function is plotted as used with $\alpha = 8$ and $\beta = 3$ to set the minimum of the function to zero. As can be seen, this loss function penalizes stronger if dark voxels in raw patches are brighter in the decoded patches. With this, the AE is forced

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to focus on reconstructing the typical appearance of vessels in abdominal ultrasound.

B. Ultrasound Data

For evaluation, a 4DUS data set (time-resolved 3DUS) is used. The data set contains liver (L) and prostate (P) scans of five different subjects (A - E) acquired under free breathing with a duration of 30 min each [19]. Due to the long-term characteristic, each scan contains a high motion-dependent variability of the anatomy. The scans were acquired on a Philips EPIQ 7 station with an X6-1 matrix transducer under robotic motion compensation [19]. Data was stored at an imaging frequency of 3 to 4 Hz.

In Fig. 4, slices of 4DUS frames of the liver and the prostate are shown. In US images of the liver, many vessels and vessel bifurcations can be identified. In the prostate volumes, the whole prostate can be imaged due to its relatively small size compared to the US field of view.

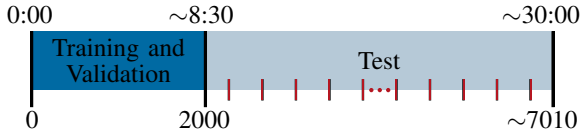


Fig. 3. Separation of the 3DUS datasets into training, validation and test sets. Points where test patches are extracted are marked in red.

The 4DUS scans are separated into training, validation and test sets. As illustrated in Fig. 3, the first 2,000 frames are used for training and validation, the remainder is used for testing. This separation is chosen due to the motivation to use the feature extractor for a tracking task where time-dependent tissue shifts are relevant. In addition, a subject- and organ-specific, as well as a general feature extractor can be tested with this separation. From the training part, 20 patches with a size of 30×30×30 voxels are randomly selected in each frame. So, a total of 40,000 patches are generated for training and validation from each subject. During training, this set is separated into 39,500 training patches and 500 validation patches. The test part contains 5,010 frames in liver scans and 6,820 frames in prostate scans on average. To cover the entire test part, a patch is chosen randomly from every 25th frame to create a patch set for testing. On average, test sets with 200 and 240 patches are created for liver and prostate, respectively. In Fig. 4, such a patch is plotted by a red square in both illustrated slices. The spatial resolution of the scans differ but the mean resolution is (1.1×0.7×1.0) mm in liver scans and (0.9×0.6×1.0) mm in prostate scans. On average, the patches represent a tissue area with a size of (30×20×40) mm.

C. Experiments

The prepared patch data sets are used to train different AEs. The AE architecture is kept fixed while the data sets used for training are varied.

Three kinds of AEs (*AE1-3*) are trained and four kinds of experiments (*E1-4*) are performed. First, ten AEs of type *AE1* are trained by using training data of one subject and one organ, so these AEs are subject- and organ-specific. For training the AEs of type *AE2*, training sets of one organ from four out of five subjects are used. Ten subject-general and organ-specific AE2s are generated. The last kind of AEs, *AE3*, are based on *AE2* but they are retrained with the training data of the subject excluded from the training phase in *AE2*. The AEs are retrained for another 25 epochs while all weights in the model are allowed to be updated. *AE3* is divided into two sub groups depending on the organ that is used for retraining. In *AE3a*, data of the same organ as in *AE2* is used while in *AE3b*, data of the other organ is used. Consequently, ten AEs are trained for types *AE3a* and *AE3b*, respectively.

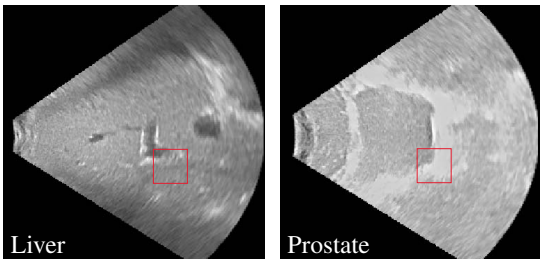


Fig. 4. Slices of 3DUS volumes of the liver and the prostate. The red squares are examples for patches with a size of 30×30×30 voxels.

The first experiment *E1* is a subject- and organ-specific test. The AEs *AE1* are applied to the test sets of the subject and the organ they were trained with. Experiment *E2* is divided into two parts, depending on whether the organ of training and test sets are the same. In experiment *E2a*, a cross-subject test is conducted for the subject- and organ-specific AEs of type *AE1*, using different subjects but the same organ as they were trained on. Similarly, experiment *E2b* performs a cross-organ test for *AE1*. The experiment *E3* is also divided into *E3a*, where the AEs of the type *AE2* are applied to the test sets with the same organ of the subject that was excluded from training, and *E3b* where the AEs of the type *AE2* are applied to test sets with the other organ. In the experiments *E4a* and *E4b*, the AEs of the type *AE3a* and *AE3b* are used, respectively. The AEs are applied to the test sets of the subjects they were retrained on. Experiment *E4a* conducts an organ-specifc test, *E4b* a cross-organ test. The descriptions of these experiments and the AEs that are trained are summarized in Tab. I and an example is given for each experiment.

Evaluating the reconstruction accuracy of the AEs can be done by comparing the input patches and the patches encoded and decoded by the AE. An ideal AE would be able to perfectly reconstruct an input patch, so no difference can be measured between the input and the output. However, due to information loss because of data compression in the AE, there will be a measurable difference. To quantify this difference,

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TABLE I
DESCRIPTION OF THE DATA USED TO TRAIN THE AES AE1-3 AND TO PERFORM THE EXPERIMENTS E1-4.

Experiments		Training		Example
E	Description	AE	Description	
E1	subject and organ specific	AE1	subject and organ specific	A-L→ A-L
E2a	cross subject, organ specific	AE1		A-L→ B-L,C-L,D-L,E-L
E2b	subject specific, cross organ	AE1		A-L→ A-P
E3a	subject general, organ specific	AE2	subject general, organ specifc	ABCD-L → E-L
E3b	subject general, cross organ	AE2		ABCD-L → E-P
E4a	subject general, organ specific	AE3a	AE2 retrained organ specific	ABCD-L + E-L → E-L
E4b	subject general, cross organ	AE3b	AE2 retrained cross organ	ABCD-L + E-P → E-P

two metrics are used. The mean absolute error (MAE)

$$\mathrm{MAE}(x,y) = \frac{1}{d}\sum_{i=1}^{d}|x_i - y_i| \tag{2}$$

quantifies the absolute difference of the voxel intensities of two patches x and $y \in \mathbb{R}^d$ where d is the number of voxels in a patch. Note that the MAE quantifies the absolute voxel intensity differences, but it is not necessarily able to measure if an input patch and a decoded patch correlate. To achieve this, the normalized cross correlation (NCC)

$$\mathrm{NCC}(x,y) = \frac{\sum_{i=1}^{d}(x_i-\bar{x})(y_i-\bar{y})}{\sqrt{\sum_{i=1}^{d}(x_i-\bar{x})^2\sum_{i=1}^{d}(y_i-\bar{y})^2}} \tag{3}$$

of two patches x and $y \in \mathbb{R}^d$ is used. Here, $\bar{x}$ and $\bar{y}$ are the mean of x and y, respectively. NCC is contrast invariant though it quantifies if the voxel intensities have a similar trend in the input and output patches. In medical applications, the anatomical structures within an image are important. By using NCC it can be verified if the AE is able to properly encode these important structures into the latent space, independent of their voxel intensities.

Fig. 5 shows that the reconstructed patches are smoothed in comparison to the original patches. US images are affected by noise and imaging artifacts like speckle that cannot be adequately learned by the AE and thus cannot be reconstructed. The missing noise leads to a difference between the original and the reconstructed patches which can be considered acceptable because no image information is lost, but it affects the reconstruction accuracy. The AE is smoothing the patches in an unknown way. To approximately quantify the difference between the original patches and smoothed patches, a 3×3×3 mean filter was applied to all patches in the test data sets. In Fig. 5, two examples are visualized. The mean NCC and MAE between the original and filtered patches is 0.94 and 4.07, respectively. These numbers represent an upper boundary for the reconstruction quality of the AE that we cannot expect to be exceeded and thus needs to be considered when interpreting the reconstruction accuracy of an AE.

The AEs *AE3b* are used to verify if it is possible to perform tracking in latent space. In each data set, the corresponding retrained AE is used for encoding ultrasound patches. Due to a missing ground truth of the tissue motion, latent space tracking is compared to a commonly used method in called template or pattern matching. In each data set, a template for tracking is chosen manually. This template is tracked for 500 consecutive frames by using template matching in latent space as well as in image space. Considering the mean imaging frequency of 3.5 Hz, the template is tracked for around 2.4 min. In a new 3DUS volume, the target template is searched by sampling the volume. In latent space tracking, the target template is found by encoding the samples and searching for the most similar sample using the L_2 norm. In image space tracking, the most similar sample is found by using sum of squared differences (SSD) as metric.

III. RESULTS AND DISCUSSION

The AEs are evaluated by applying them to the test data sets and quantifying the similarity of the input and output patches. First, in experiment *E1* the subject- and organ-specific AEs *AE1* are applied to the corresponding test sets to determine a baseline. The NCC values of the baselines are in a range of 0.81-0.85 for liver patches and 0.64-0.81 for prostate patches. In Fig. 5, two original patches and their reconstructions from the experiments *E1-4* of the liver and prostate are visualized. The reconstructions of *E1* have a high similarity to the original patches.

In *E2a*, cross-subject and cross-organ tests are performed. Fig. 6 and 7 show that both the cross-subject experiment *E2a* and the cross-organ experiment *E2b* lead to an NCC decrease and an MAE increase with respect to the baseline. This could be explained by inter-subject variability, even though all 4DUS sets of one organ cover the same anatomical region, vessels and other fine structures differ between subjects. An AE trained on one subject cannot reconstruct patches of another subject with a high accuracy. The variability in a given data set used for training is too small to transfer the AE to another subject or organ. Fig. 5 illustrates that in reconstructions of the experiments *E2a* and *E2b* rough structures of the original patches can be identified but the AEs could not recover finer structures.

In the experiment *E3*, the AEs were trained by data of one organ from different subjects. In Fig. 6, we again see that NCC and MAE increase and decrease, respectively, when compared to experiment *E2*. While the NCC and MAE values for the liver in the experiment *E3a* are getting closer to the baseline,

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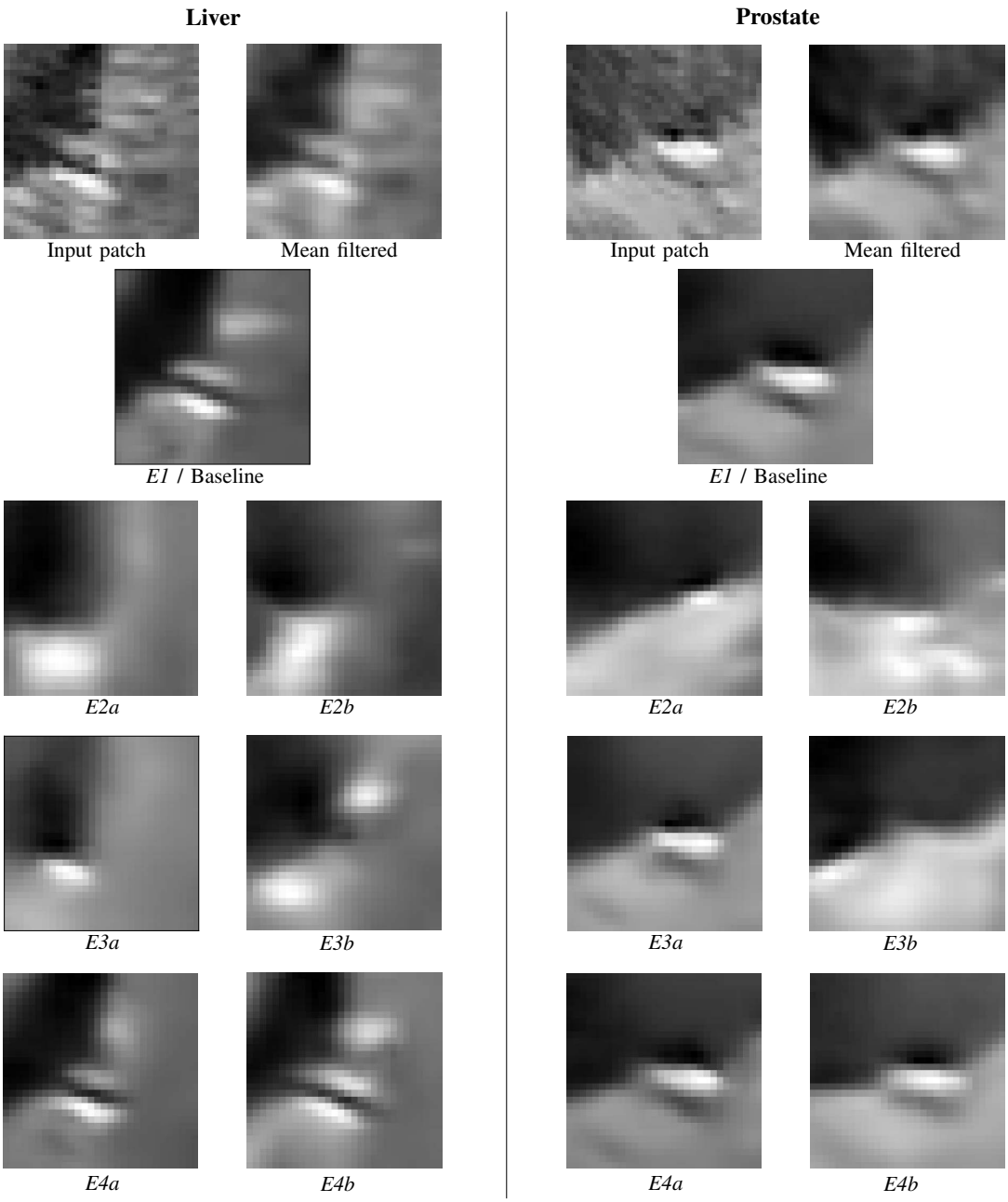


Fig. 5. Comparison between slices of original and reconstructed patches of the liver (left) and prostate (right) depending on the experiments *E1-4*. Reconstructed patches of subject depending (left sub-blocks) as well as organ depending (right sub-blocks) AEs are presented.

the results for the prostate even outperform the baseline. The structure variability between subjects in prostate US images is lower than in liver images. Based on the data of four different subjects, the prostate-specific AEs were able to reconstruct patches of an unknown subject with high accuracy, in contrast to the liver-specific AEs. Thus, an organ-specific AE trained on multiple subjects performs with a higher accuracy than the highly individual baseline, trained only on one subject. Due to the higher structural variability, liver-specific AEs likely need a larger amount of different subjects to outperform the baseline with a subject-general AE. This indicates that it is feasible to train a subject general AE if sufficient training data is available.

The experiments *E4* show that retraining can enhance the reconstruction accuracy if the AE has not been pretrained on a sufficient amount of data. In the organ-specific experiment *E4a*, the accuracy of the AEs trained and applied on the liver increases and reaches the baseline, while accuracy of the AEs on the prostate remains constant when compared to experiment *E3a*. These AEs could not be improved by retraining. In

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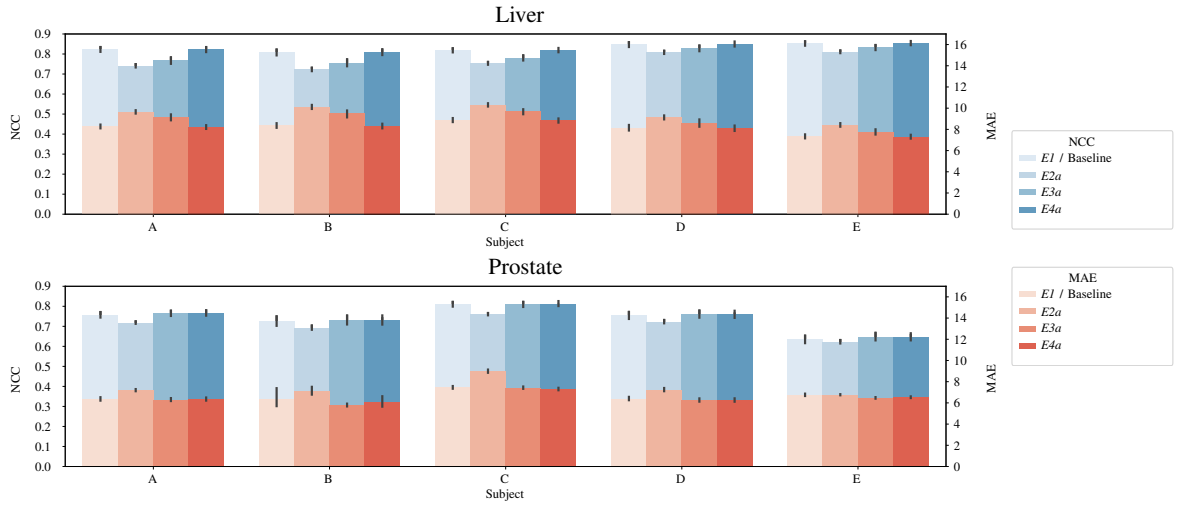


Fig. 6. Mean NCC (blue) and MAE (red) of subject-dependent AE tests on the liver (top) and prostate (bottom) test patches.

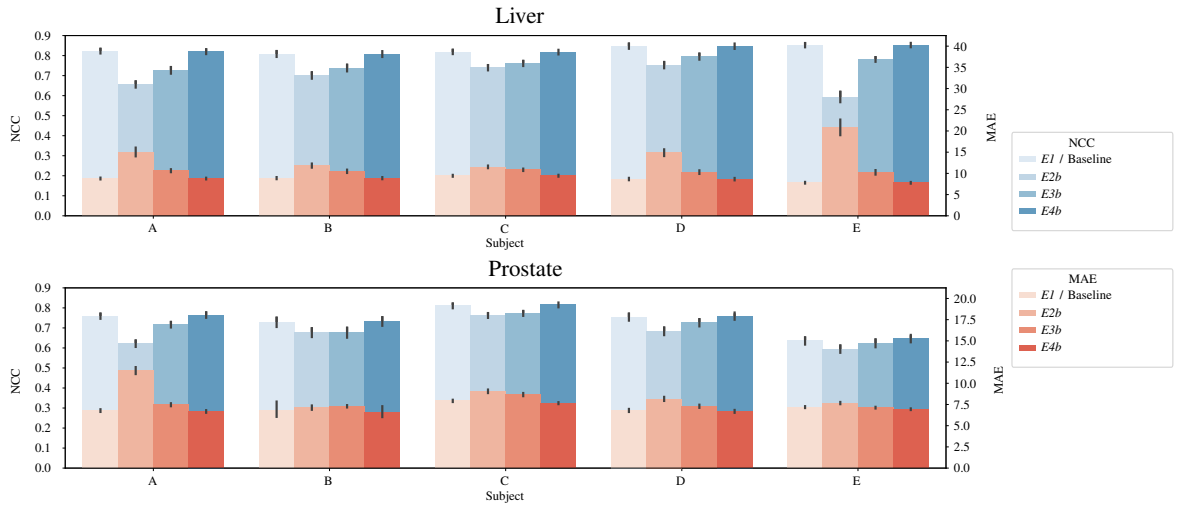


Fig. 7. Mean NCC (blue) and MAE (red) of organ-dependent AE tests on the liver (top) and prostate (bottom) test patches.

addition, experiment *E4b* shows that the accuracy of all AEs increases with respect to experiment *E3* and the baseline is nearly reached. Retraining an AE that has been pretrained on another organ strongly improves the accuracy as can be seen in Fig. 5.

The results show that there is no appreciable difference between retraining an AE that was pretrained on the same or on a different organ. By retraining on subject- and organ-specific data the AE's accuracy is strongly increased. This can be used, for example, in image-guided interventions like radiation therapy where a subject-specific treatment plan is prepared. Furthermore, if enough training data is available to overcome the organ-specific subject variability, it is possible to train an AE that does not have to be retrained (*E4a*, prostate). These AEs are useful for organ-specific or more generalized diagnostic applications as they can be applied without retraining.

The ultrasound data used in this study is acquired with one ultrasound station and similar device settings for each data set. However, in practice a multitude of different ultrasound stations and different device settings can be used. This inter-observer variability affects the ultrasound images. Though the device settings differ slightly between the ultrasound data sets, inter-observer variability is not considered in this study but will be investigated in the future.

By using the AEs *AE3b* a first latent space tracking feasibility study is performed. The positions of the most similar samples in latent space as well as in image space tracking are compared to each other. In Fig. 8, such a comparison is visualized along the three primary volume axes in data of subject A. Note that the discrete trajectory is caused by the discrete voxels in the patches and the motion amplitude at the y-axis is smaller than at the other axes.

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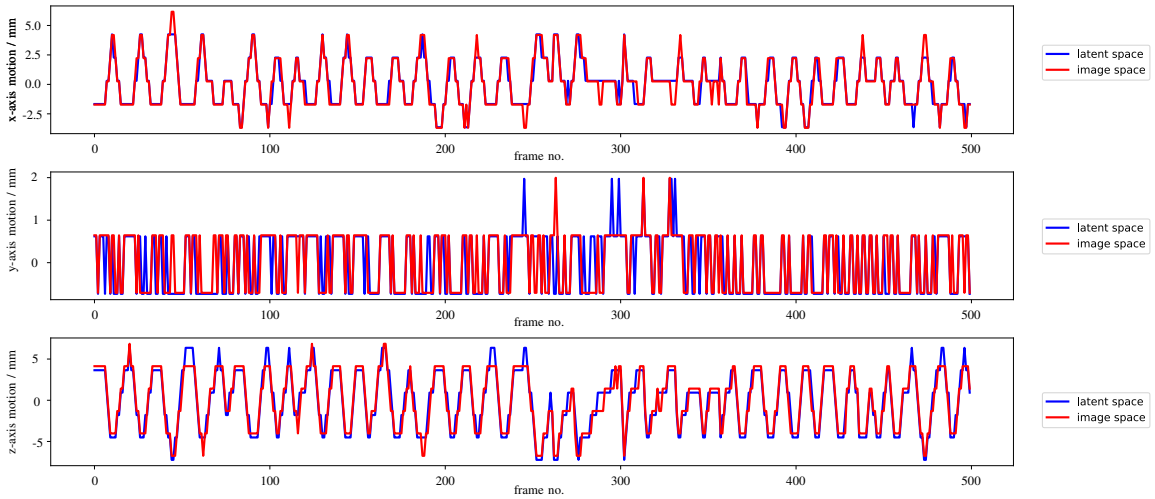


Fig. 8. Example for the target motion tracked by latent space (red) and image space tracking (blue) along the three volume axes in liver data of subject A. The step-like pattern is caused by the discrete voxel size of the image data.

The trajectories of latent space and image space tracking positions are similar. All experiments show trajectories with sinusoidal patterns that can be attributed to motion due to breathing. This indicates a successful template tracking. To quantify the similarity, the mean euclidean distances between the positions of latent space and image space tracking are determined and presented in Tab. II. The differences are in the millimeter range for the liver and even reach sub-millimeter in the prostate. Considering the mean 3DUS resolution of (1.1×0.7×1.0) mm in liver scans and (0.9×0.6×1.0) mm in prostate scans, the differences are in the voxel-range. The mean positional differences in the liver are higher compared to the prostate, especially for subjects B, D and E. This can likely be attributed to the smaller motion range of the prostate in contrast to the liver. The liver is moving with respiration, causing translation, rotation and even deformation of the organ and its internal structures, thus changing their appearance in ultrasound images. This makes continuous tracking more challenging compared to more stable organs such as the prostate.

TABLE II
MEAN EUCLIDEAN DISTANCES BETWEEN LATENT SPACE AND IMAGE SPACE TRACKING POSITIONS

Organ	Subject	Mean euclidean distance / mm
liver	A	0.6±1.1
	B	4.7±5.7
	C	0.7±1.0
	D	7.9±4.9
	E	3.7±2.5
prostate	A	0.6±0.8
	B	0.4±0.9
	C	0.2±0.5
	D	0.1±0.3
	E	0.5±0.6

Due to the lack of ground truth regarding tissue motion, it was not possible to obtain reliable information about the tracking accuracy, but the results indicate that tracking in latent space by using an AE is able to achieve similar results as tracking by using template matching in image space. It should be noted that this tracking study is a first feasibility study so the method is not optimized. Furthermore, the utilized templates were chosen manually so there is no information about the suitability of the templates for tracking. Templates that are easy to track for an observer are not necessarily easy to track for an algorithm. Further studies are necessary to investigate the selection of the optimal templates for tracking.

IV. Conclusion

The aim of this work was to determine the feasibility of generating a general AE that is able to encode 3DUS patches of different organs and subjects and thus serve as a new feature extractor. We could show that it is possible to train specific as well as general AEs to map 3DUS patches into a small latent space. The AEs are able to encode 3DUS patches into 0.2 % of their original size and reconstruct them with high accuracy, showing that the most important image information could indeed be stored in the latent space. Pretrained AEs that were retrained by currently used data achieved the highest reconstruction accuracy, though there was nearly no difference between organ-specific or cross-organ pretraining. An AE that is pretrained on a multitude of different 3DUS patches can be retrained for a specific task and potentially be used for feature extraction in different tasks such as segmentation, detection or classification as well as for motion tracking. Such a general AE could simplify these tasks by replacing complex feature extraction methods.

A major drawback for training this general AE is the large amount of data required from different subjects and organs. In future work, more 3DUS data will be acquired to train and improve the general AE. Furthermore, the inter-observer

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variability will be investigated to create an observer-general AE.

Furthermore, we performed a first feasibility study to check the plausibility of the template trajectory induced by latent space tracking. In future work, the developed tracking method will be investigated in detail and improved to potentially outperform more traditional methods such as template matching. In addition, a ground truth will be generated to be able to measure the tracking accuracy. This initial feasibility study indicates that tracking in latent space after encoding patches is a promising method for target tracking in abdominal ultrasound.

ACKNOWLEDGMENT

Many thanks go to Ivo Kuhlemann, who recorded and provided the 3DUS data.

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